

# INDIFFERENCE CURVE ANALYSIS

## **Section A**

Duration of each question is **40 mins**.

- 1** With the help of an indifference curve diagram, assess the extent to which a rise in price would affect the demand for a normal good differently from the demand for a Giffen good. [20]
  
- 2** Evaluate the use of indifference curve analysis to derive the demand curve for a normal good and the demand curve for an inferior good. [20]

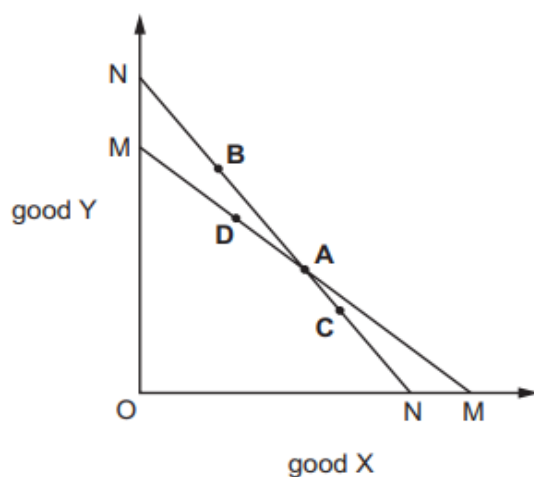
## Section B

Answer **all** questions

Duration 1hr 15mins

- The diagram shows budget lines of a consumer choosing between two goods, X and Y. Initially the budget line is MM and the consumer's preferred position is at point A. Subsequently the money prices of both goods change shifting the budget line to NN.

Which point could represent the preferred position of the consumer after the change in prices if their tastes remained unchanged?

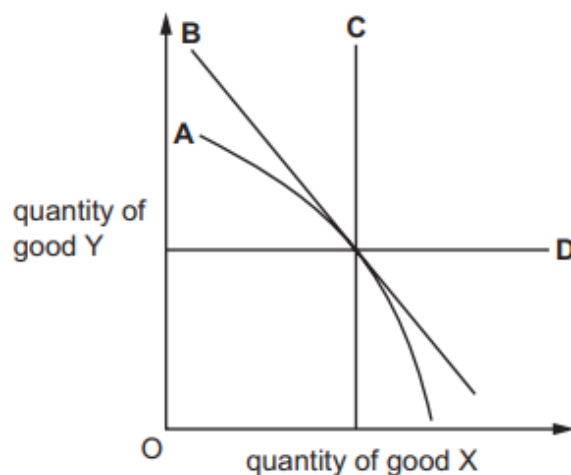


- What is equal at all points along an indifference curve for an individual consumer?

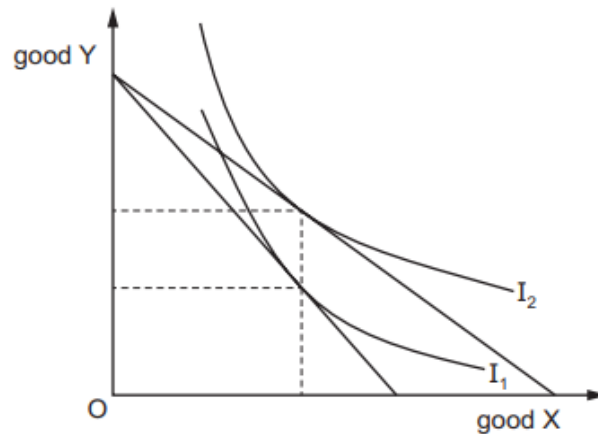
- A** marginal expenditure
- B** marginal utility
- C** total expenditure
- D** total utility

- The diagram shows four unusual indifference curves.

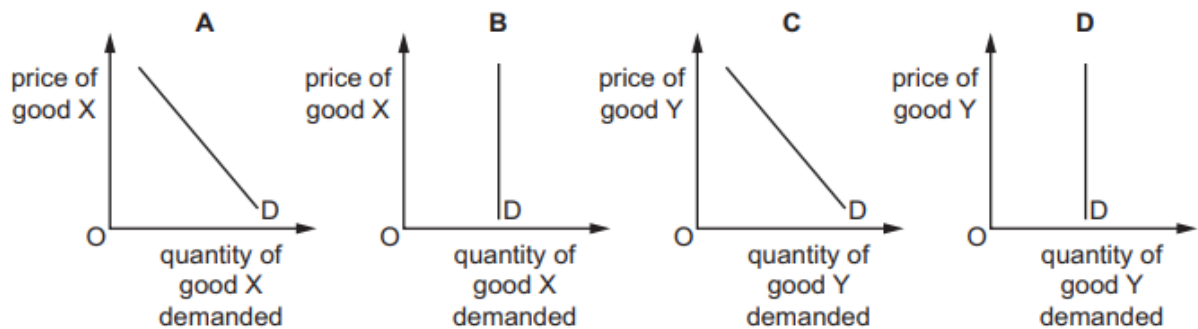
Which curve indicates that the goods are perfect substitutes for the consumer?



4. A consumer demands two goods, X and Y. The indifference curve diagram shows the effect of a change in the price of one of these goods on the consumer's equilibrium.

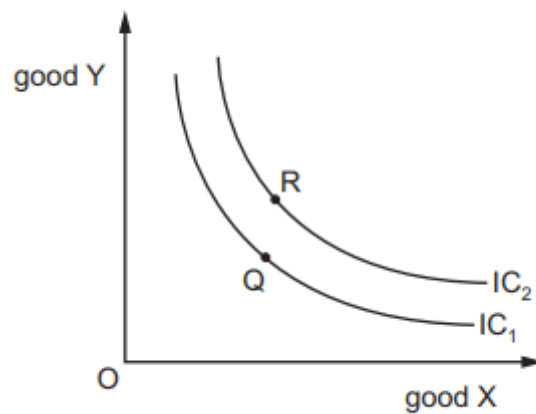


Which demand curve can be deduced from the diagram?



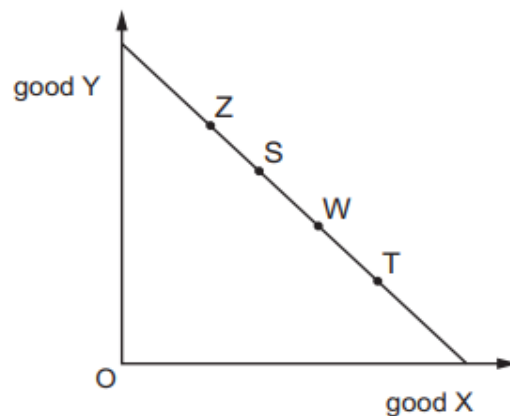
5. Which statement about indifference curves is correct?
- A** The indifference curves for inferior goods can show higher satisfaction on lower curves.
  - B** The indifference curves for normal goods show a diminishing marginal rate of substitution.
  - C** The indifference curves for perfect complements go through the origin.
  - D** The indifference curves for perfect substitutes are right-angled.

6. The diagram shows two indifference curves for a consumer.



What can be concluded if the consumer's equilibrium moves from Q to R?

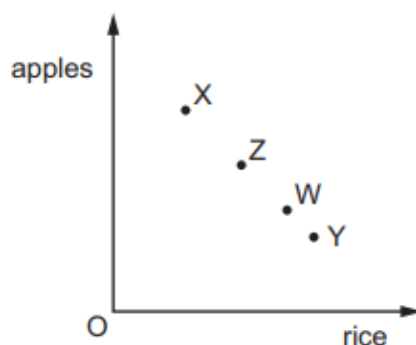
- A The consumer is acting rationally.
  - B The consumer's money income is unchanged.
  - C The opportunity cost of good Y is constant.
  - D The price of good X has risen.
7. The diagram shows a budget line indicating combinations of good Y and good X that a consumer could buy with a given income. The combinations shown by points S and T both lie on the same normal indifference curve.



Which statement is logically correct?

- A The consumer is indifferent between combinations W, S and T.
- B The consumer prefers combination T to combination W.
- C The consumer prefers combination W to combination S.
- D The consumer prefers combination Z to combination T.

8. When does the slope of a budget line always change?
- A A person buys more of one of the two goods.
  - B A person's income changes.
  - C One of the two goods becomes more desirable.
  - D The price of one of the two goods changes.
9. What would **not** affect the budget line of an individual consumer?
- A the individual's preference for various goods
  - B the level of income tax
  - C the money prices of goods
  - D the wages earned by the individual
10. The diagram shows various combinations of apples and rice which are potentially available to a consumer.



If standard indifference curves were added to the diagram they would show that the consumer is indifferent between combination X and combination W.

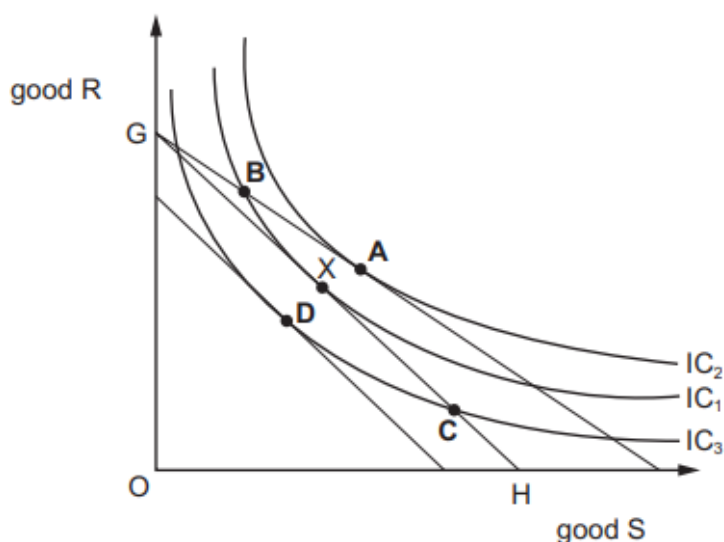
What can be concluded?

- A The consumer is indifferent between combination Y and combination Z.
- B The consumer is indifferent between combination Z and combination W.
- C The consumer prefers combination Y to combination X.
- D The consumer prefers combination Z to combination W.

11. The diagram shows a consumer's initial budget line is GH and a set of indifference curves  $IC_1$ ,  $IC_2$  and  $IC_3$  for goods R and S. The original equilibrium for the consumer is point X.

The inflation rate is rising faster than money incomes.

What will be the most likely new equilibrium for the consumer if income is spent?

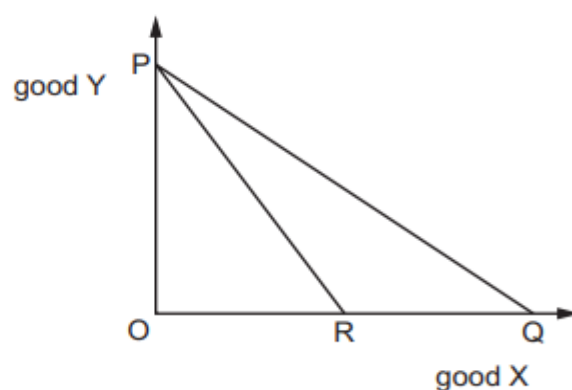


12. A consumer can purchase two goods, X and Y, from a given income.

What will cause the consumer's budget line to make a parallel shift to the right?

|          | consumer's income | price of good X | price of good Y |
|----------|-------------------|-----------------|-----------------|
| <b>A</b> | 10% decrease      | 20% decrease    | 10% decrease    |
| <b>B</b> | 10% decrease      | 20% decrease    | 20% decrease    |
| <b>C</b> | 10% increase      | 10% increase    | 20% decrease    |
| <b>D</b> | 10% increase      | 20% increase    | 20% increase    |

13. The diagram shows budget lines for normal goods X and Y.

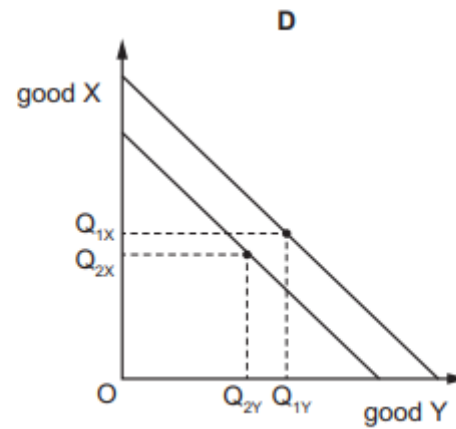
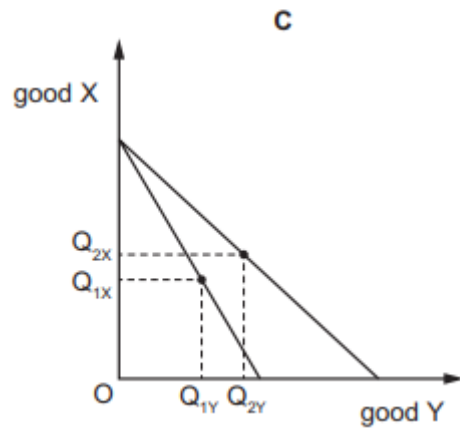
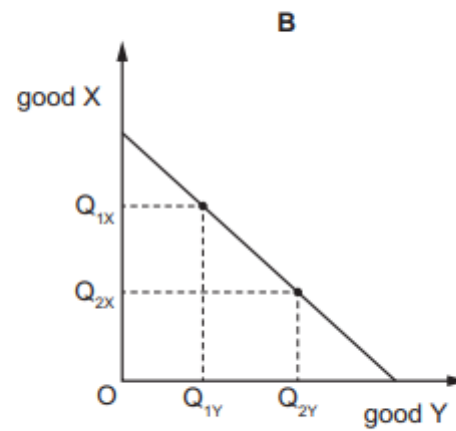
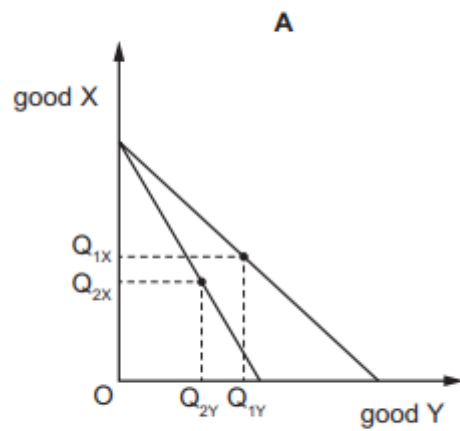


What could cause a budget line to shift from PQ to PR?

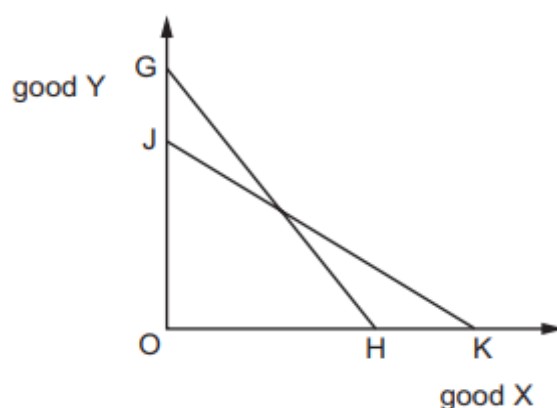
|          | price of X | price of Y | level of consumer incomes |
|----------|------------|------------|---------------------------|
| <b>A</b> | fall       | rise       | fall                      |
| <b>B</b> | no change  | rise       | no change                 |
| <b>C</b> | rise       | no change  | no change                 |
| <b>D</b> | rise       | no change  | rise                      |

14. The diagrams show different budget lines and consumer equilibrium choices. The consumer is in equilibrium at  $Q_{1X}Q_{1Y}$ .

Which diagram shows the effect of a successful advertising campaign of good Y?



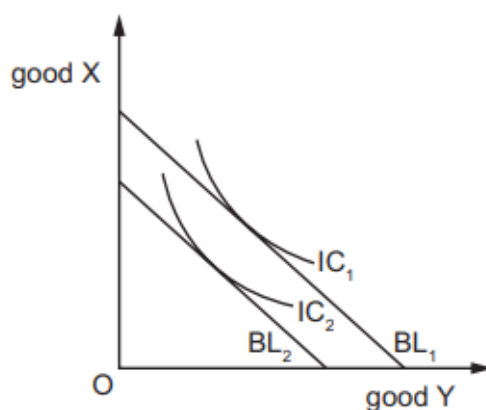
15. The curve JK in the diagram is a consumer's initial budget line.



Which combination could cause the budget line to shift to GH?

|          | price of good X | consumer's money income |
|----------|-----------------|-------------------------|
| <b>A</b> | increase        | increase                |
| <b>B</b> | decrease        | decrease                |
| <b>C</b> | increase        | decrease                |
| <b>D</b> | decrease        | increase                |

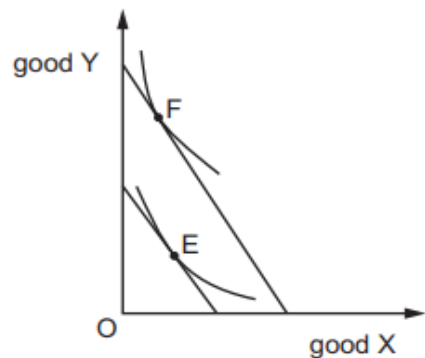
16. The diagram shows the indifference curves (IC) for a consumer for goods X and Y together with the consumer's budget lines (BL).



If the budget line shifts from  $BL_1$  to  $BL_2$ , how does this affect the consumer's preferences for goods X and Y when prices remain the same?

- A** He continues to buy the same amounts of X and Y.
- B** He prefers to buy fewer of X and fewer of Y.
- C** He prefers to buy fewer of X and more of Y.
- D** He prefers to buy more of X and fewer of Y.

17. The indifference curve diagram shows the effect of an increase in income on the consumption of good X and good Y.

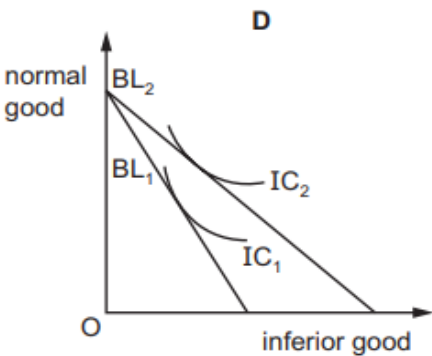
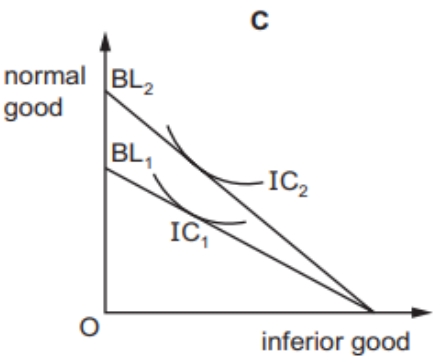
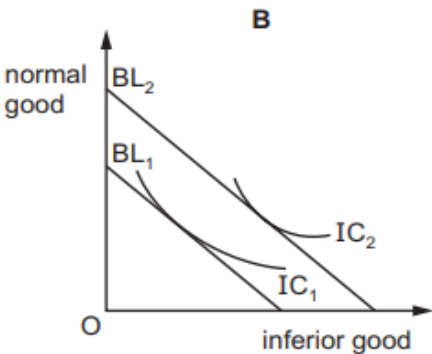
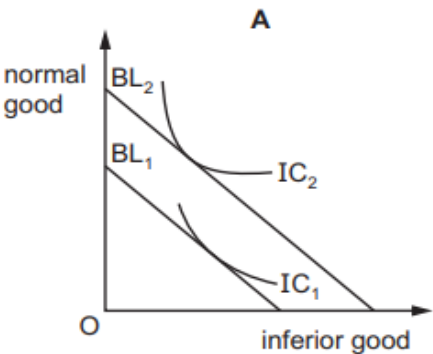


Which types of good X and good Y are indicated by the change in consumption from E to F?

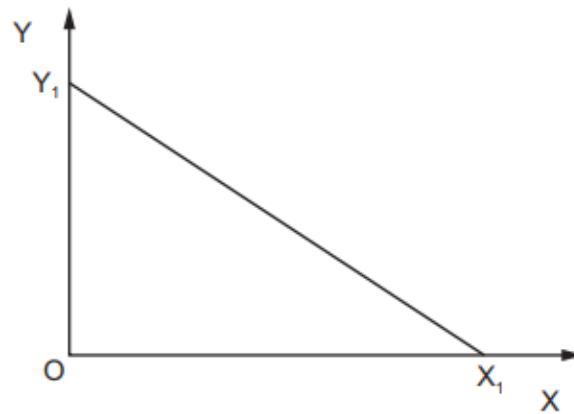
|          | good X   | good Y   |
|----------|----------|----------|
| <b>A</b> | inferior | inferior |
| <b>B</b> | inferior | normal   |
| <b>C</b> | normal   | inferior |
| <b>D</b> | normal   | normal   |

18. The diagrams show a consumer's indifference curves (IC) and budget lines (BL) for an inferior good and a normal good.

Which diagram shows the effect of a cut in income tax on the consumer's choice?

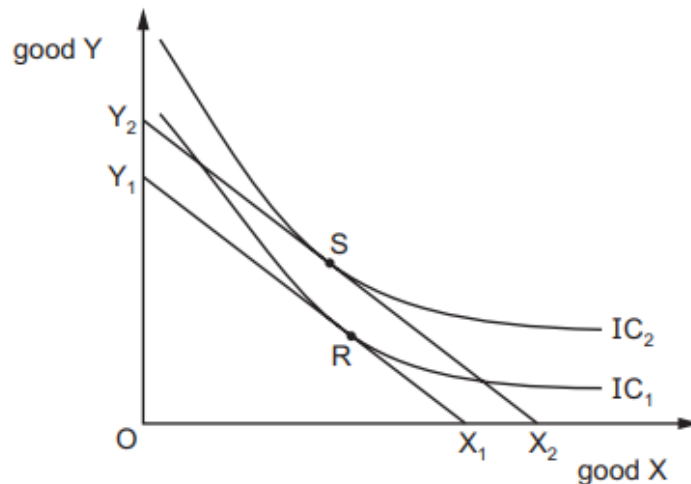


19. The diagram below shows a customer's budget line for two goods, X and Y.



Assuming the price of X does not change, the slope of the budget line will become steeper when:

- A the consumer's income falls.
  - B the consumer's income rises.
  - C the price of Y falls.
  - D the price of Y rises.
20. The diagram shows two indifference curves and two budget lines for goods X and Y. The consumer's initial equilibrium is point R.

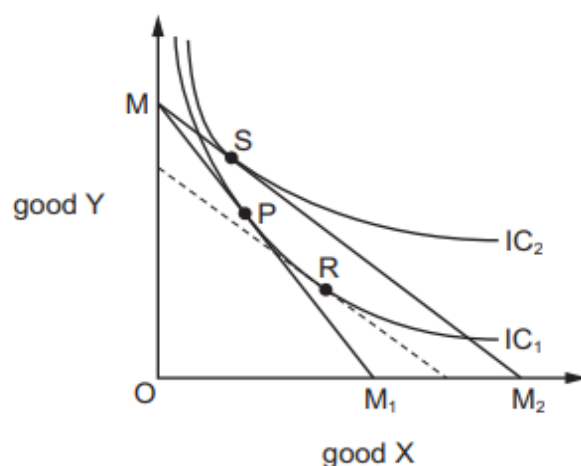


There is a shift in the budget line from  $Y_1X_1$  to  $Y_2X_2$  and the new equilibrium for the consumer is point S.

What can definitely be concluded about good X?

- A It is a Giffen good.
- B It is an inferior good.
- C It is a normal good.
- D It is a public good.

21. The diagram shows two indifference curves and two budget lines for two goods X and Y.

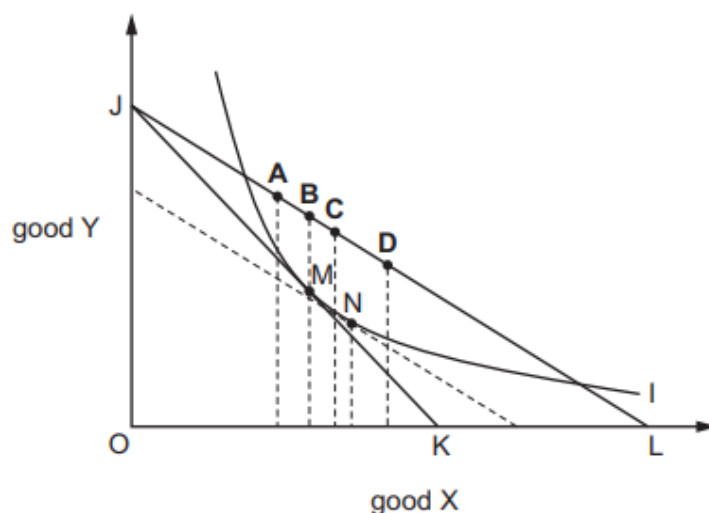


The initial position is P. P to R is a substitution effect. R to S is an income effect.

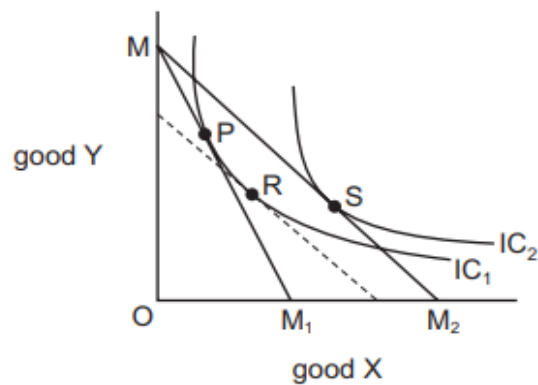
What type of good is good X?

- A** a Giffen good
  - B** a luxury good
  - C** a normal good but not a luxury good
  - D** an inferior good but not a Giffen good
22. In the indifference curve diagram point M is the consumer's initial equilibrium and MN is the substitution effect of a fall in the price of good X.

If good X is a Giffen good which point will be the consumer's new equilibrium point after the fall in the price of good X?



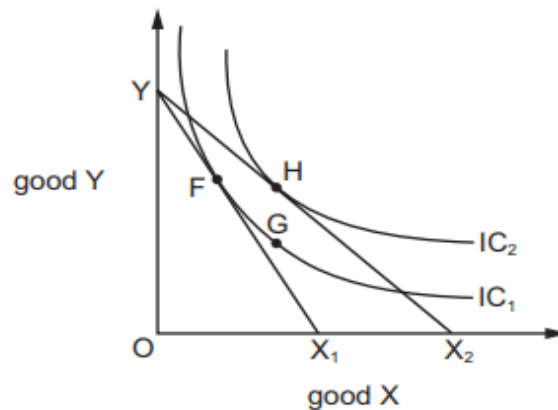
23. The diagram shows two indifference curves and two budget lines for two goods X and Y.



The initial position is P. P-R is a substitution effect. R-S is an income effect.

What type of good is good X?

- A a Giffen good
  - B a luxury good
  - C a normal good
  - D an inferior good
24. The diagram shows two indifference curves and two budget lines for goods X and Y.



The consumer's preferred final position becomes point H.

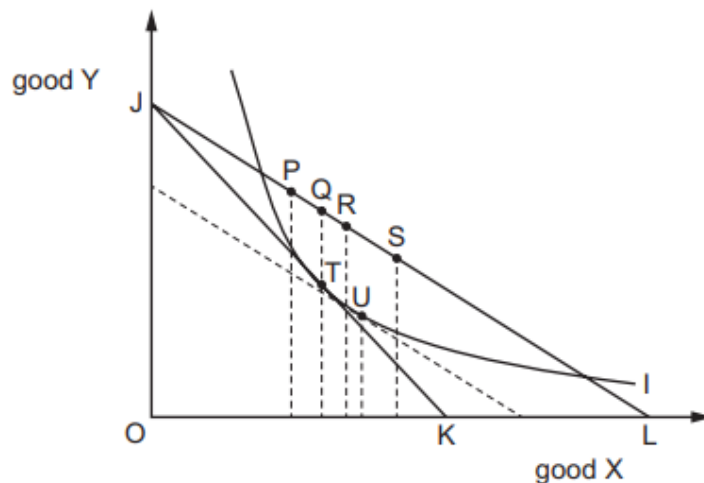
What does the movement from G to H represent?

- A the income effect of a price fall for X
- B the price effect of a price change for X
- C the substitution effect of a price fall for X
- D the substitution effect of a price rise for X

25. When the price of a good falls the effect on the quantity demanded is the result of an income effect and a substitution effect.

Which statement about these effects is correct?

- A For inferior goods the income effect and the substitution effect work in the same direction.
  - B For inferior, but not Giffen, goods the income effect outweighs the substitution effect.
  - C For normal goods the income effect and substitution effect work in the same direction.
  - D For normal goods the income effect outweighs the substitution effect.
26. What is **not** held constant when calculating the income effect of a change in the price of a good?
- A the consumer's money income
  - B the consumer's preferences
  - C the consumer's real income
  - D the prices of other goods
27. The diagram shows budget lines and an indifference curve. The consumer's initial position is T. The price of good X then falls.



Which combination of statements about the movements in this diagram is correct?

- A T to Q represents perfect elasticity; T to P represents a Giffen good
- B T to R represents an income effect; R to S represents a substitution effect
- C T to U represents a substitution effect; T to Q represents a normal good
- D T to U represents a substitution effect; U to P represents an income effect

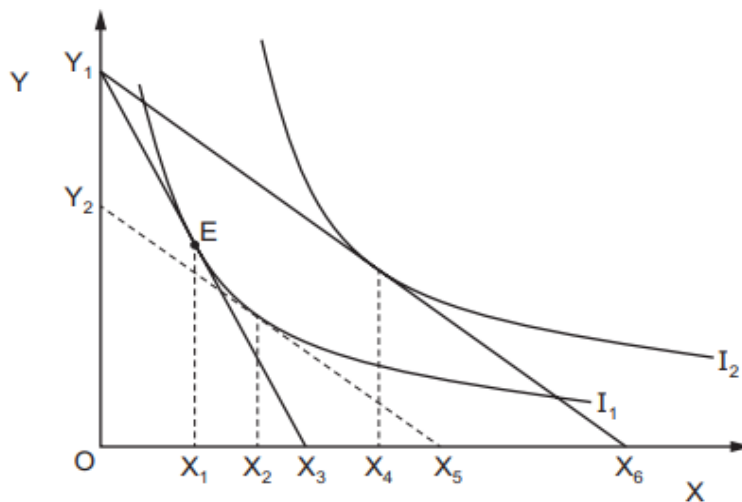
28. The table shows the substitution effect and income effect for a Giffen good and an inferior good as the price of the good changes.

Which combination is correct?

|          | type of good | price change | demand change from substitution effect | demand change from income effect |
|----------|--------------|--------------|----------------------------------------|----------------------------------|
| <b>A</b> | Giffen       | fall         | positive                               | negative                         |
| <b>B</b> | Giffen       | rise         | positive                               | positive                         |
| <b>C</b> | inferior     | fall         | negative                               | negative                         |
| <b>D</b> | inferior     | rise         | negative                               | negative                         |

29. A consumer spends all of their income on two goods, Y and X, and is at position E. The price of X falls and the price of Y remains constant.

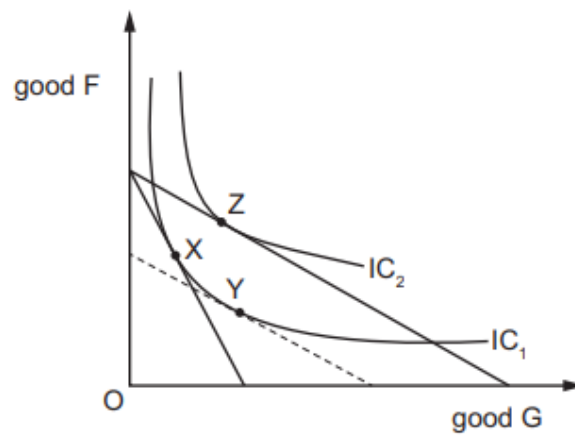
The graph shows indifference curves and budget lines which are used to determine the price, income and substitution effects that are related to this price change.



Which distance gives the income effect of this price change?

- A**  $X_1X_2$       **B**  $X_1X_4$       **C**  $X_2X_4$       **D**  $X_5X_6$

30. The diagram shows a consumer's indifference curves (IC) for goods F and G together with the consumer's budget lines.



What could explain the movement from X to Z on the diagram?

- A** a fall in the price of G when G is an inferior good
- B** a fall in the price of G when F is a Giffen good
- C** a fall in the price of G when G is a normal good
- D** a fall in the price of G when G is a Giffen good

INDIFFERENCE CURVE  
ANALYSIS  
(MARKING SCHEME)

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1        | <p><b>With the help of an indifference curve diagram, assess the extent to which a rise in price would affect the demand for a normal good differently from the demand for a Giffen good.</b></p> <p>Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question.</p> <p>AO1 and AO2 out of 14 marks.<br/>AO3 out of 6 marks.</p> <p><b>Indicative content</b><br/>Responses may include:</p> <p><b>AO1 Knowledge and understanding and AO2 Analysis</b></p> <ul style="list-style-type: none"> <li>• Diagram showing Indifference curves, budget lines and price lines.</li> <li>• Changes in budget lines representing a rise in price.</li> <li>• Diagrams of resultant path of consumption showing differences in substitution and income effects for normal and Giffen goods.</li> <li>• Comment on changes in overall demand based on changes in income and substitution effects.</li> <li>• Up to Level 2 only if no diagram.</li> </ul> <p><b>AO3 Evaluation (max 6 marks)</b></p> <ul style="list-style-type: none"> <li>• Comment on extent of changes would vary depending on the price elasticity of demand a</li> <li>• It would also depend on the slope of the indifference curve and the marginal rate of substitution.</li> </ul> | 20    |
|          | AO1 Knowledge and understanding and AO2 Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 14    |
|          | AO3 Evaluation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 6     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |
| 2        | <p><b>Evaluate the use of indifference curve analysis to derive the demand curve for a normal good and the demand curve for an inferior good.</b></p> <p>Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question.</p> <p>AO1 and AO2 out of 14 marks. AO3 out of 6 marks.</p> <p><b>Indicative content</b> Responses may include:</p> <p><b>AO1 Knowledge and understanding and AO2 Analysis</b></p> <ul style="list-style-type: none"> <li>Assumptions of indifference curve analysis including:</li> <li>rational consumers who prefer more to less,</li> <li>a two-good world,</li> <li>the prices of the goods are (initially) constant,</li> <li>the goods can be substituted for each other,</li> <li>transitivity, if there is indifference between A and B and indifference between B and C, there is indifference between A and C</li> <li>perfect knowledge of the market.</li> <li>Definition of an indifference curve (IC), explanation of convex IC and analysis of rising levels of satisfaction and the creation of an IC map.</li> <li>Definition of a budget line (BL) and analysis of rising/falling income levels on the BL and consumption</li> <li>Explanation of the substitution effect (SE) and the income effect (YE) when the price of a good changes.</li> <li>Analysis of the effect of a positive and negative YE when combined with the SE to account for a normal and inferior good.</li> <li>The combination of the IC and BL maps to show point of tangency between IC and BL to give a combination of the 2 goods that gives the highest level of satisfaction given the level of income.</li> </ul> <p><b>AO3 Evaluation</b></p> <ul style="list-style-type: none"> <li>Consumer rationality: indifference analysis assumes that consumers act rationally. They are of a calculating mind, carrying numerous combinations of different commodities in their heads, can substitute one for the other, compare their total utilities and make a rational choice between various combinations of goods.</li> </ul> | 20    |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2        | <ul style="list-style-type: none"> <li>As the effect of a reduction in price is an increase in demand for both a normal and inferior good (except in the case of Giffen good) it is not possible to know whether a good is normal or inferior.</li> <li>It is difficult to apply the concept of substitutability to goods such as consumable durables, which are one-off indivisible purchases and the choice is between, for example, 1 fridge and 4 weeks' food.</li> <li>A consumer is faced with a much more complex world than the two-good world of the model because a consumer buys not just two but a large number of commodities to satisfy their innumerable wants.</li> <li>IC analysis is a static analysis and consumer preferences may change over time.</li> <li>Accept all valid responses.</li> </ul> |       |
|          | AO1 Knowledge and understanding and AO2 Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 14    |
|          | AO3 Evaluation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 6     |

# INDIFFERENCE CURVE ANALYSIS

## A LEVEL

### ECONOMICS P3

|     |   |     |   |     |   |
|-----|---|-----|---|-----|---|
| 1.  | B | 11. | D | 21. | A |
| 2.  | D | 12. | B | 22. | A |
| 3.  | B | 13. | C | 23. | C |
| 4.  | B | 14. | B | 24. | A |
| 5.  | B | 15. | A | 25. | C |
| 6.  | A | 16. | B | 26. | C |
| 7.  | C | 17. | B | 27. | D |
| 8.  | D | 18. | A | 28. | A |
| 9.  | A | 19. | C | 29. | C |
| 10. | D | 20. | B | 30. | A |